CS3003: IT Project Management

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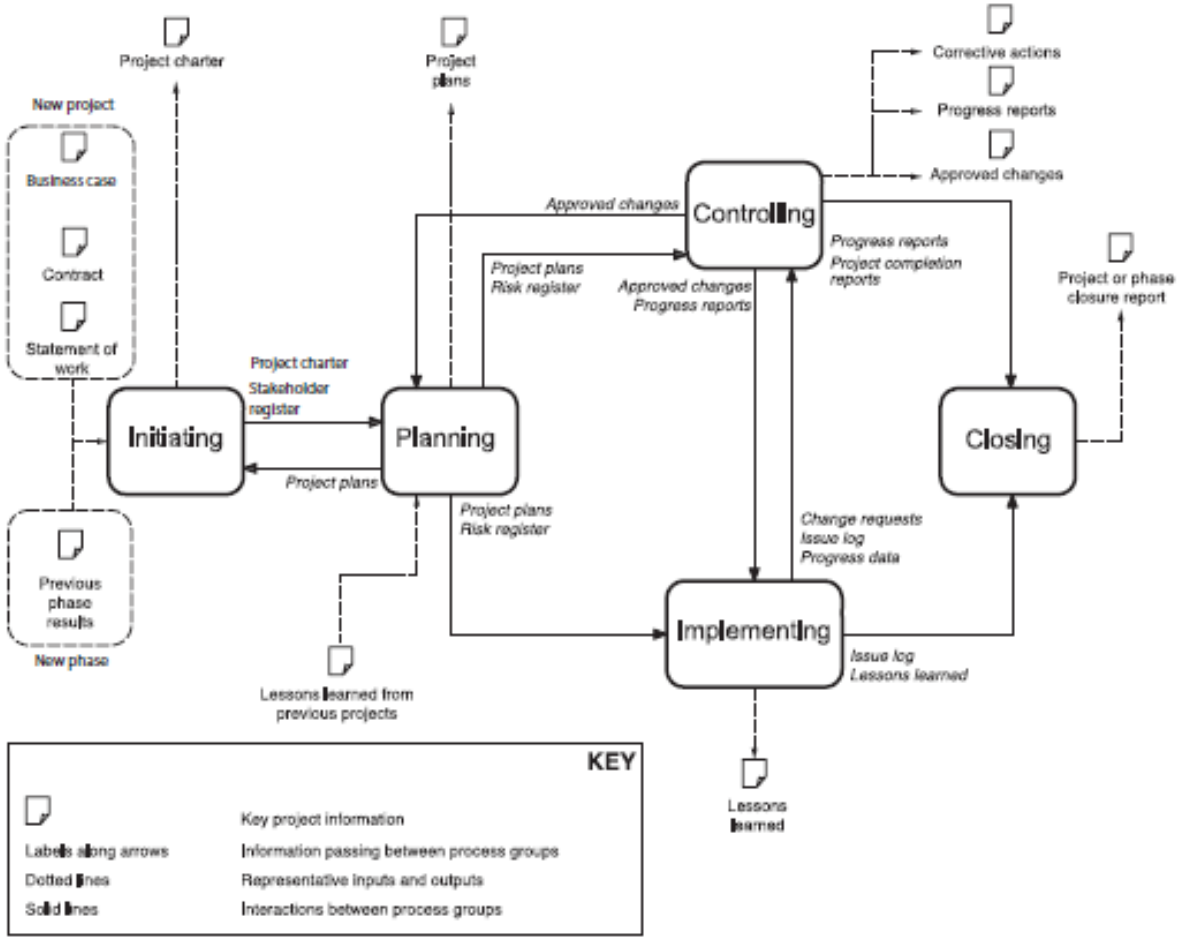
Part 4: The first/basic Project Planning Elements: Scope, Time, Resources, Communications.

Project Planning Overview and the Focus of Part 4 (see items in red)

Subject groups	Process groups: Initiating	Planning	Implementing	Controlling	Closing
Integration	4.3.2 Develop project charter	4.3.3 Develop project plans	4.3.4 Direct project work	4.3.5 Control project work 4.3.6 Control changes	4.3.7 Close project phase or project 4.3.8 Collect lessons learned
Stakeholder	4.3.9 Identify stakeholders		4.3.10 Manage stakeholders		
Scope		4.3.11 Define scope 4.3.12 Create work breakdown structure 4.3.13 Define activities		4.3.14 Control scope	
Resource	4.3.15 Establish project team	4.3.16 Estimate resources 4.3.17 Define project organization	4.3.18 Develop project team	4.3.19 Control resources 4.3.20 Manage project team	
Time		4.3.21 Sequence activities 4.3.22 Estimate activity durations 4.3.23 Develop schedule		4.3.24 Control schedule	
Cost		4.3.25 Estimate costs 4.3.26 Develop budget		4.3.27 Control costs	
Risk		4.3.28 Identify risks 4.3.29 Assess risks	4.3.30 Treat risks	4.3.31 Control risks	
Quality		4.3.32 Plan quality	4.3.33 Perform quality assurance	4.3.34 Perform quality control	
Procurement		4.3.35 Plan procurements	4.3.36 Select suppliers	4.3.37 Administer procurements	
Communication		4.3.38 Plan communications	4.3.39 Distribute information	4.3.40 Manage communications	

NOTE The purpose of this table is not to specify a chronological order for carrying out the activities. Its purpose is to map subject groups and process groups.



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Scope: Requirements and Work Breakdown Structure (WBS)

Scope Management Overview

1. Define Scope Statement & Requirements

↓

2. Create Work Breakdown Structure

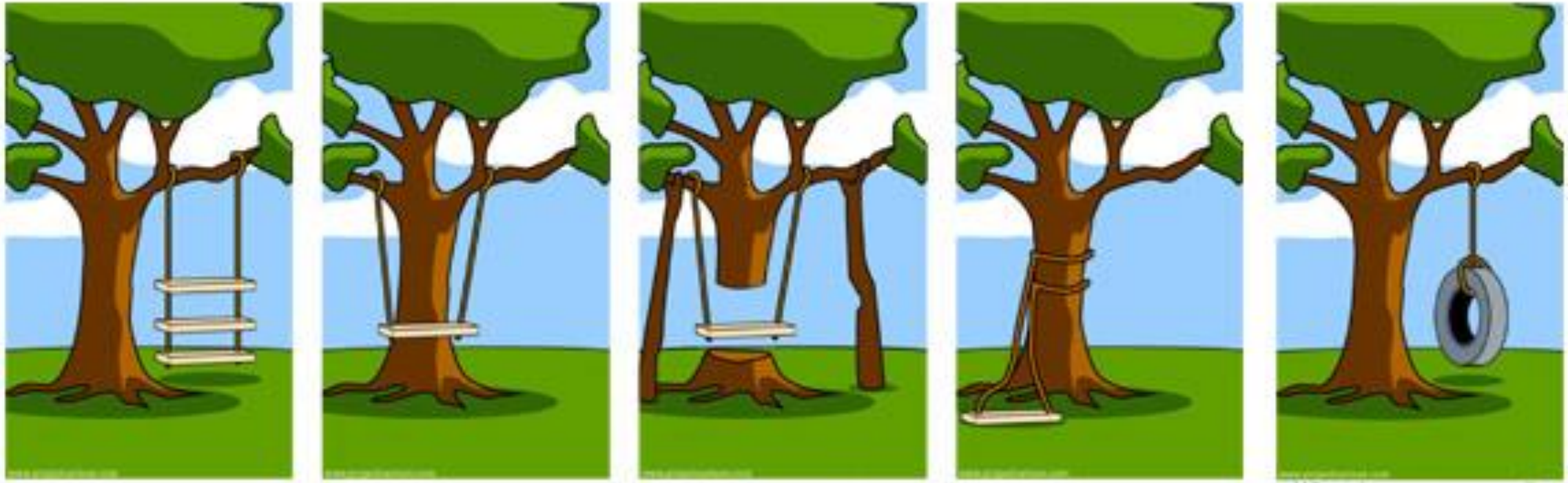
↓

3. Define Activities
(produce an Activity List)

Scope Management: Requirements (brief Intro)

- Functional Vs non-Functional
- A number of Issues
  - Stakeholders don’t know what they really want.
  - Stakeholders express requirements in their own terms.
  - Different stakeholders may have conflicting requirements.
  - Organisational and political factors may influence the system requirements.
  - The requirements change during the analysis process. New stakeholders may emerge and the business environment may change.
  - Clarity and Unambiguity, Common Communication language, amalgamation (several different requirements may be expressed together)

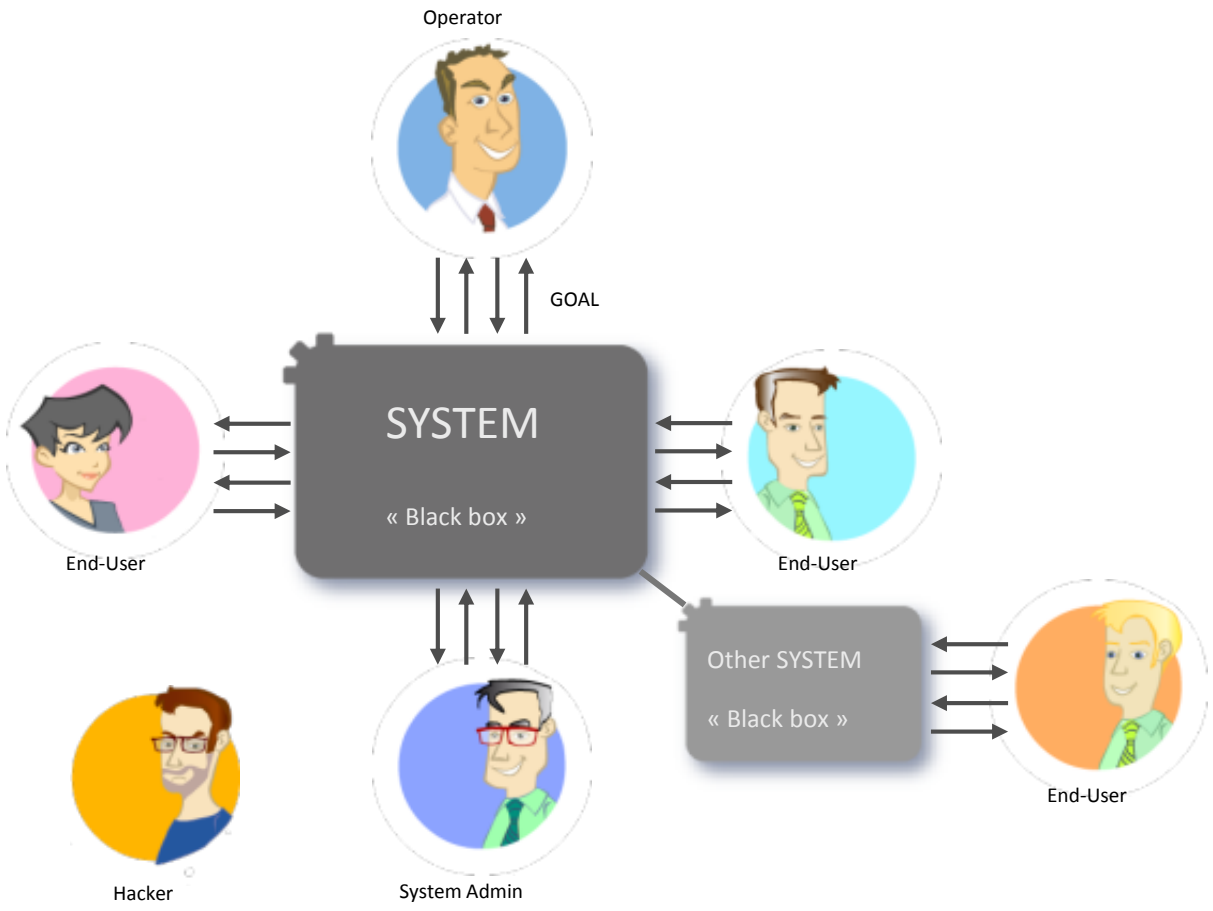
Scope Management: Requirements (brief intro)



Scope Management: Requirements Document (sample ToC)

Chapter	Description
Preface	This should define the expected readership of the document and describe its version history, including a rationale for the creation of a new version and a summary of the changes made in each version.
Introduction	This should describe the need for the system. It should briefly describe the system's functions and explain how it will work with other systems. It should also describe how the system fits into the overall business or strategic objectives of the organization.
Glossary	This should define the technical terms used in the document. No assumptions should be made about the experience or expertise of the reader.
User requirements definition	Here, you describe the services provided for the user. The nonfunctional system requirements should also be described in this section. This description may use natural language, diagrams, or other notations that are understandable to customers. Product and process standards that must be followed should be specified.
System architecture	This chapter should present a high-level overview of the anticipated system architecture, showing the distribution of functions across system modules. Architectural components that are reused should be highlighted.
System requirements specification	This should describe the functional and nonfunctional requirements in more detail. If necessary, further detail may also be added to the nonfunctional requirements. Interfaces to other systems may be defined.
System models	This might include graphical system models showing the relationships between the system components and the system and its environment. Examples of possible models are object models, data-flow models, or semantic data models.
System evolution	This should describe the fundamental assumptions on which the system is based, and any anticipated changes due to hardware evolution, changing user needs, and so on. This section is useful for system designers as it may help them avoid design decisions that would constrain likely future changes to the system.
Appendices	These should provide detailed, specific information that is related to the application being developed; for example, hardware and database descriptions. Hardware requirements define the minimal and optimal configurations for the system. Database requirements define the logical organization of the data used by the system and the relationships between data.

Scope Management: Requirements and Context Diagram



Scope Management: Main Ways of Writing a Requirements *Specification*

Notation	Description
Natural language	The requirements are written using numbered sentences in natural language. Each sentence should express one requirement.
Structured natural language	The requirements are written in natural language on a standard form or template. Each field provides information about an aspect of the requirement.
Graphical notations	Graphical models, supplemented by text annotations, are used to define the functional requirements for the system; UML use case and sequence diagrams are commonly used.
Mathematical specifications	These notations are based on mathematical concepts such as finite-state machines or sets. Although these unambiguous specifications can reduce the ambiguity in a requirements document, most customers don't understand a formal specification. They are also time-consuming.

Scope Management: Some Methods of Requirements Elicitation

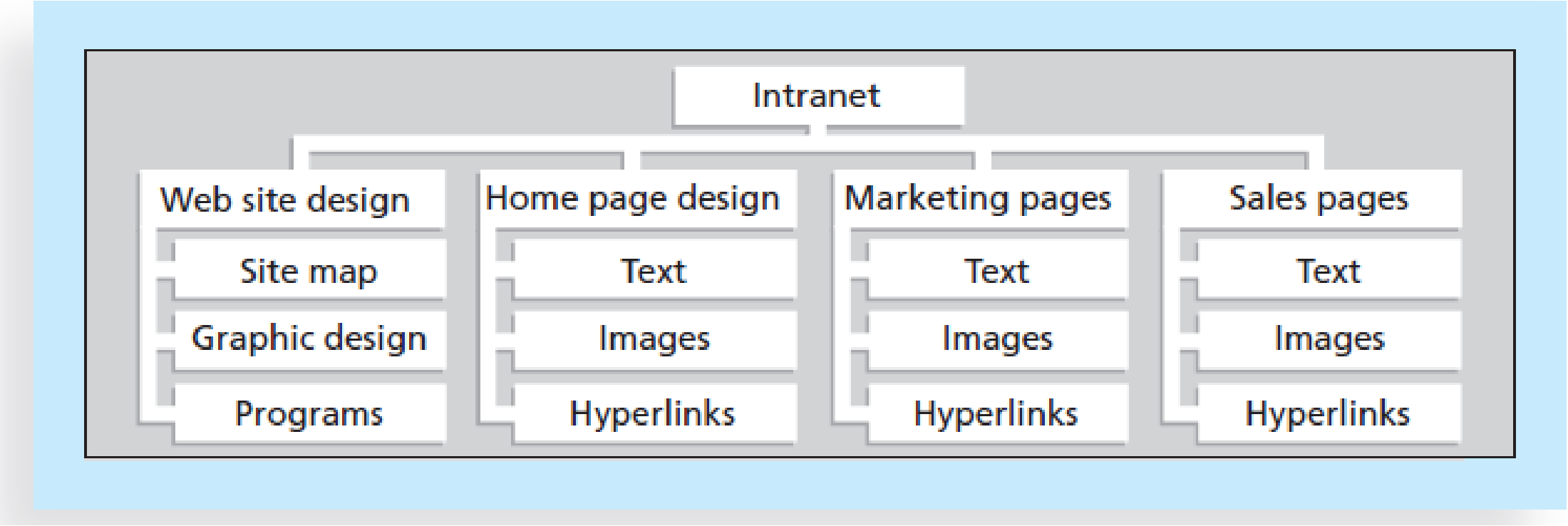
1. Interviews
2. Brainstorming workshops
3. Questionnaires and Surveys
4. Ethnography
5. Use Case Analysis, User Stories
6. Prototypes/Modelling
7. Demonstrations of current operating environment
8. 5-Why Analysis
9. Fishbone diagrams
10. Use experience / examples / blueprints from previous projects

Scope Management: Work Breakdown Structure (WBS)

- WBS is a “deliverable-oriented grouping of the work involved in a project that defines its total scope” (PMBoK). Can be a hierarchy of deliverables or activities/tasks/work items. The goal is to decompose what needs to be done (task) in a project into smaller chunks.
- Teams may organize the WBS around
  i. **project products** or
  ii. **project phases** or
  iii. **project management process groups**.
- In “traditional” PM, WBS is very important as it provides the basis for project schedule, costs, resources planning and change management.
- WBS typically have an accompanying **WBS Dictionary** – a document that describes/provides details on each of the elements of the WBS (e.g. a more detailed description, start-end dates, quality criteria/requirements, required resources, applicable technical content, applicable costs)

What is a similar concept in Agile/Scrum approaches? What is the main difference?

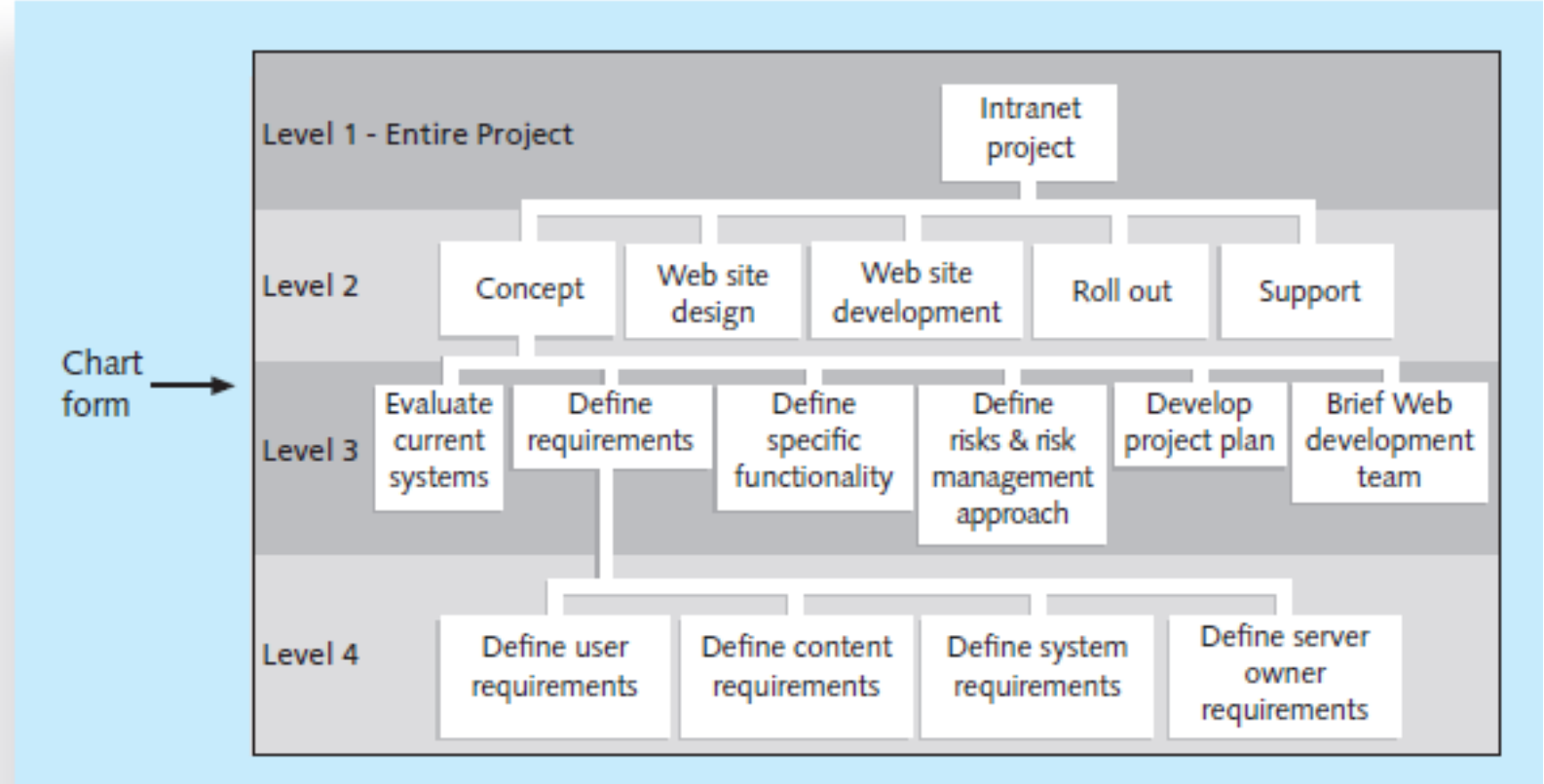
Scope Management: Work Breakdown Structure (WBS) Examples



Sample intranet WBS organized by product

Image from Schwalbe's book .

Scope Management: Work Breakdown Structure (WBS) Examples



Sample intranet WBS organized by phase

Tabular form with PMI numbering

1.1 Concept
- 1.1.1 Evaluate current systems
- 1.1.2 Define requirements
  - 1.1.2.1 Define user requirements
  - 1.1.2.2 Define content requirements
  - 1.1.2.3 Define system requirements
  - 1.1.2.4 Define server owner requirements
- 1.1.3 Define specific functionality
- 1.1.4 Define risks and risk management approach
- 1.1.5 Develop project plan
- 1.1.6 Brief Web development team

1.2 Web site design
1.3 Web site development
1.4 Roll out
1.5 Support

Images from Schwalbe's book .

Scope Management: Product Breakdown Structure (PBS)

In **Prince2**, the concept of **Product Breakdown Structure** (PBS) is *explicitly* used.

PBS is defined as "*a hierarchical structure that breaks down a final product into its constituent sub-products*". The sub-products include any Prince2 management artefacts to be produced.

The PBS is usually accompanied by a *Product Flow Diagram (PFD)* that defines the sequence in which the products will be developed.

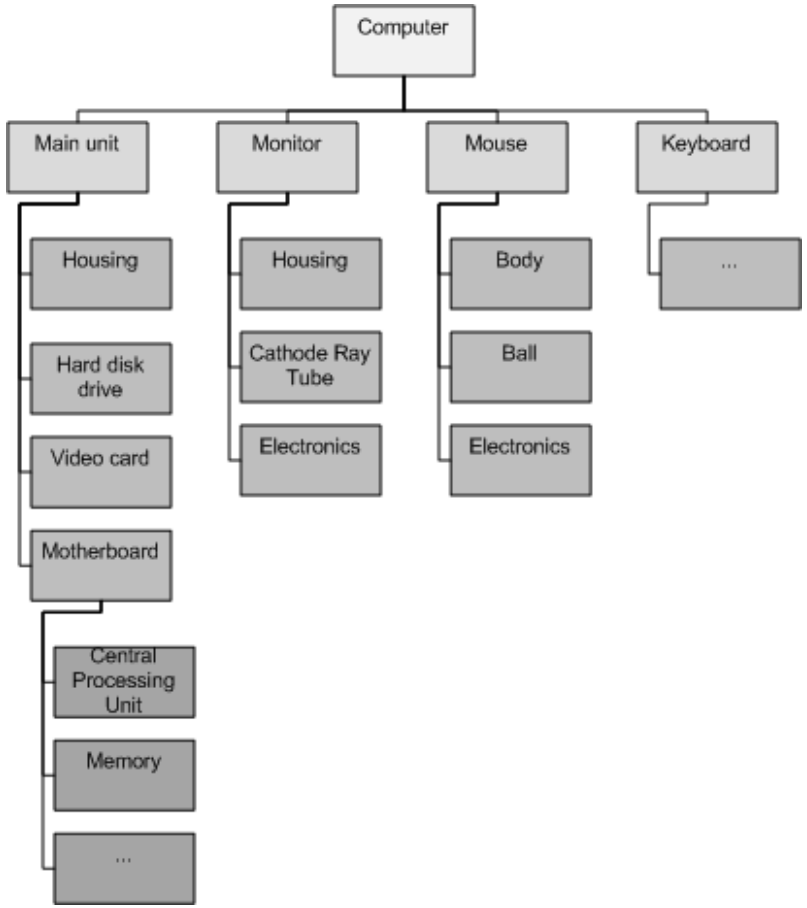


Image from Wikipedia .

Scope Management: IBM's MITP methodology for WPS

Level 1. *Project*

Level 2. *Deliverables* (e.g. SW, HW, Manuals, Training)

Level 3. *Components to produce deliverables* (e.g. modules but also items such as tests)

Level 4. *Work-Packages,* i.e. major work items or collections of related *tasks* required to produce a component

Level 5. *Fine-grained Tasks* (single person responsibility)

Image from Wikipedia .

Scope Management: Work Breakdown Structure (WBS)

Main approaches to develop a WBS/PBS	
Analogous	Reuse/adapt a WBS of a previous project
Top-Down approach	Start with largest items and break them into smaller ones, i.e. in each box, one asks the Question: “What tasks need to be done in order to complete this piece of work?” The answer is written in boxes one level down… The process is repeated for subsequent levels
Bottom-Up approach	Identify as many tasks as possible. Then aggregate and organize into summary activities in higher levels of the WBS
Mind Mapping	No lists or structure of tasks, a non-linear structure. Good approach for the development of a PBS